



Corrosion inhibition of mild steel in hydrochloric acid environment using thiadiazole derivative: Weight loss, thermodynamics, adsorption and computational investigations

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ABSTRACT

The corrosion inhibition potency of 1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO) for mild steel in 1.0 M hydrochloric acid was investigated by applying weight loss techniques, and quantum chemical calculations. Conditions that determine the potential for corrosion inhibition, including; concentration, the structure of ETO, and solution temperature, were chosen for the study. The highest inhibitive efficacy of ETO was 98.4%, at the optimum concentration (500 ppm) and room temperature. ETO has superior corrosion inhibitive potency in the HCl environment, owing to the presence of the thiadiazole and benzene rings, in addition to carbonyl and amino groups. Applicable thermodynamical equations were used to calculate the activation energy, enthalpy, and entropy. The adsorption isotherms were applied to determine the Gibbs free energy difference. The experimental findings of the investigation revealed that the activation energy of an inhibited process was higher than for an uninhibited process. Furthermore, the increased inhibition efficiency with improving temperature, and the values of ΔG° , indicated that ETO molecules coated the mild steel surface in both chemical adsorption and physical interactions. The adsorption process on the mild steel surface obeys the Langmuir adsorption isotherm. Both experimental and density functional theory (DFT) findings in the current investigation are in excellent agreement.

1. Introduction

Corrosion inhibition has received a lot of attention and research, since it is one of the most efficient and cost-effective ways to preserve materials (Al-Amiery et al., 2014). Several studies and reports have been published on potent organic inhibitors (with heteroatoms including oxygen, nitrogen, and sulfur) as well as pi-bonds and/or aromatic rings in their molecular structures (Ebenso et al., 2010a; Al-Taweel et al., 2020; El-Belghiti et al., 2016).

Due to its outstanding mechanical and chemical properties, mild steel is widely used in a variety of technical and manufacturing applications, including design and construction. However, one of the biggest drawbacks of using mild steel is its corrosive nature (Ebenso et al.,

2010b). Mild steel corrosion is an inevitable but controllable phenomenon. Annealing is one of the most important processes to prevent corrosion of mild steel.

Acid pickling is a systematic industrial cleaning process used to get rid of chemical oxides and material deposits in the petrochemical industry and in oil well activities (Masoud et al., 2010). Due to the extremely harmful corrosive effect of the metallic acid used (HCl), this process must be controlled. To meet the environmental requirements, the use of environmentally friendly organic inhibitors arose. As a result, inorganic inhibitors (including chromates, molybdates, phosphates, and nitrates) are widely used in metal/alloy corrosion protection, due to their high efficacy (Khaled and Amin, 2009). Organic inhibitors, in addition to being environmentally friendly, also limit corrosion. They

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have high electron density centers (N, O, S, P and pi bonds), which allows them to adsorb on metal surfaces and protect them from corrosive environments (Mert et al., 2014; Wang et al., 2007; Khaled, 2007). Compounds containing heteroatoms such as O, N, S and P, which produce physical/chemical adsorption on metal surfaces, have been reported to be effective inhibitors (Fu et al., 2012).

To gain a better understanding of how organic compounds suppress corrosion, it is important to investigate the adsorption mechanism of these molecules on the metal surface. The efficacy of most organic corrosion inhibitors is typically related to their adsorption capabilities at the metal/solution interface. This may be investigated using adsorption isotherms which can reveal more about the adsorption mechanism as well as the types of interactions that occur between the inhibitors and the steel surface. Indeed, corrosion inhibitors can adsorb at the interface via two types of interactions: physical adsorption (weak interactions), resulting from electrostatic interactions between charged molecules and the metal surface, and chemical adsorption, resulting from electron sharing between inhibitors and d-orbitals of the iron surface. The inhibitory activity of these chemicals improves in the order $O < N < S < P$ (Vosta and Eliasek, 1971).

The authors investigated the corrosion inhibitory behavior of thiazides, specifically 1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO), see Fig. 1, on corrosive mild steel in HCl solution. This is a continuation of previous work on the screening of organic inhibitors containing N and O (Al-Amiery, 2021a; Al-Amiery et al., 2020a; Al-Baghdadi et al., 2021a; Al-Amiery et al., 2020b; Eltmimi et al., 2021a; Alamiery et al., 2021a). Apart from its ability to chelate with metals, the lone pairs of electrons on heteroatoms (N, O and S) and the presence of pi bonds in the entire structure, are believed to be major factors that enhance the adsorption of this molecule on a mild steel surface. Corrosion inhibition, thermodynamics, adsorption coefficients, and theoretical studies of 1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO) for mild steel in 1.0 M HCl media have been investigated.

The novelty of this work is the study of the protective response of ETO, which was analyzed analytically on the corrosion of mild steel in 1.0 M hydrochloric acid using gravimetric and thermometric techniques. The main objective was to investigate the corrosion process of mild steel in 1.0 M hydrochloric acid solution in the absence and presence of different concentrations of novel corrosion inhibitor. Another objective was to test the gravimetric studies and theoretical investigations, in addition to the thermodynamics and adsorption studies. Consequently, this research focuses on the corrosion inhibition, thermodynamics, and adsorption properties, as well as DFT investigations of ETO on mild steel in acidic solution.

2. Materials and procedures

2.1. Corrodents, materials, and chemicals

The samples for this project. The mild steel's constituent composition is shown in Table 1. The sample preparation procedure followed the ASTM G1–03 guidelines (ASTM International 2011). The strips used in weight loss procedures were cut into 2.0 cm × 0.05 cm × 1.0 cm size.

The chemicals used to synthesize the investigated inhibitor were analytical grade and could be used without further purification. They

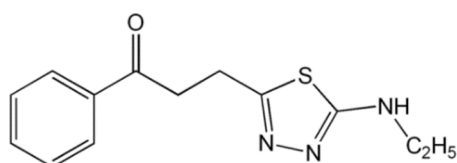


Fig. 1. The structure of 1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO).

Table 1

Mild steel chemical composition.

Elements	Fe	C	Si	P	S	Mn	Al
% wt	99.21	0.21	0.38	0.09	0.05	0.05	0.01

were obtained from Sigma Aldrich and/or Merck. The acidic environment was 1.0 M hydrochloric acid medium, prepared by diluting 37% concentrated hydrochloric acid with double distilled water.

2.2. Weight-Loss measurements

The weight-loss experiments were conducted outside in the open air. HCl (1 M) was used as a corrosive medium in the presence and the absence of varying inhibitor concentrations. Samples were placed in a water bath with a thermometer that was kept at 303° Fahrenheit. Mild steel samples (2.0 cm × 0.05 cm × 1.0 cm) were suspended in the test media for 1, 5, 10, and 24 h in triplicate. ETO concentrations of 100, 200, 300, 400, 500, and 600 ppm were investigated. Following the immersion times, the samples were extracted and submitted to the ASTM standard's post-treatment protocols G1–03 (ASTM International, 2011).

Corrosion rate (C_R) was determined using the average weight loss (mg) based on Eq. (1) (ASTM International, 2011).

$$C_R(\text{mm/year}) = \frac{87.600 \times W}{\rho a t} \quad (1)$$

where W represents the weight loss (mg), ρ is the mild steel specimen density (7.86 g/cm³), t is the exposure period, and a is the specimen surface area.

The inhibitive efficacy ($IE\%$) of ETO was determined according to Eq. (2).

$$IE\% = \frac{W_0 - W}{W_0} \times 100 \quad (2)$$

where W_0 refers to specimen weight loss without ETO and W is the specimen weight loss in presence of ETO in a corrosive environment.

To ensure the accuracy of the weight loss results, mild steel samples were weighed three times

2.3. Temperature and kinetic variables effects

Weight loss techniques were used to calculate activation energy (E_a), activation enthalpy (ΔH), and activation entropy (ΔS) at the examined temperatures (303, 313, 323, and 333 K) using the rate of corrosion and protection efficiency. Arrhenius Eq. (3) was used to calculate E_a , whereas the transition state equation was used to calculate ΔH and ΔS (Prabhu and Rao, 2019):

$$C_R = \frac{-E_a}{2.303RT} + \log K \quad (3)$$

E_a represents the activation energy, K is the exponential parameter, R refers to the universal gas constant and T is the temperature.

2.4. Factors affecting adsorption and thermodynamics

ΔH_{ads}^0 and ΔS_{ads}^0 can be determined according to Eq. (4), which is Arrhenius Alternative Equation:

$$\log \frac{C_R}{T} = \frac{-\Delta H}{2.303R} \left(\frac{1}{T} \right) + \left[\log \frac{R}{Nh} + \left(\frac{\Delta S}{2.303R} \right) \right] \quad (4)$$

where N and h are Avogadro's number and Planck's constant, respectively.

ΔH was calculated from the slope $\frac{-\Delta H}{2.303R}$ of the plot of $\log \frac{C_R}{T}$ against $\frac{1}{T}$

and ΔS was calculated from the intercept $[\log \frac{R}{N_h} + (\frac{-\Delta S}{2.303R})]$.

The adsorption isotherm describes the contact between the inhibitor molecule and the metal surface. The correlation coefficient (R²) that best fits the gathered investigative data is used to determine the isotherm. Different adsorption models, such as Langmuir, Tempkin and Freundlich adsorption isotherms, demonstrated a linear form according to Eqs. (5)–(7),

$$\text{Langmuir; } \frac{C_R}{\theta} = \frac{1}{K_{ads}} + C_R \quad (5)$$

$$\text{Temkin; } \theta = \ln C_R + K_{ads} \quad (6)$$

$$\text{Freundlich; } \log \theta = \log K_{ads} + n \log C_R \quad (7)$$

Gibb's free energy can be evaluated according to Equation (8) and was utilized to study the adsorption nature and feasibility.

$$\Delta G_{ads}^o = -RT \ln(55.5 K_{ads}) \quad (8)$$

where K_{ads} represents the constant of adsorption equilibrium and 55.5 refers to the molar water concentration.

2.5. Methodology of computation

The Gaussian 09 program Jamil et al., 2016; Frisch et al., 2009; Al-Amiry et al., 2014) was used to perform quantum chemical calculations. The B3LYP/3–21 G approach was used to optimize the structure of the tested compound in the gas phase. Quantum variables were also identified at the same theoretical level, Eqs. (9)–(12).

$$\Delta E = E_{LUMO} - E_{HOMO} \quad (9)$$

$$\eta = \Delta \frac{\Delta E}{2} \quad (10)$$

$$S = \frac{1}{\eta} \quad (11)$$

$$\mu = -\chi = \frac{E_{HOMO} + E_{LUMO}}{2} \quad (12)$$

Eq. (13) can be used to compute the fractional number of electrons transferred, ΔN , from a tested inhibitor molecule with a high potential to one with a low potential (Abdallah et al., 2012).

$$\Delta N = \frac{\chi_{Fe} - \chi_{inh}}{2(\eta_{Fe} + \eta_{inh})} \quad (13)$$

The absolute electronegativity values of the metal and inhibitor are χ_{Fe} and χ_{inh} , respectively. The absolute hardness values of the metal and inhibitor are η_{Fe} and η_{inh} , respectively.

$\chi_{Fe} = 7$ eV, $\eta_{Fe} = 0$ eV, and ΔN , in the case of iron, can be represented as Eq. (14)

$$\Delta N = \frac{7 - \chi_{inh}}{2\eta_{inh}} \quad (14)$$

3. Results and discussion

3.1. Acidic solution

Pickling of mild steel structures is usually carried out using 1.0 M HCl. To prevent the side effects of the acid, inhibitor is introduced. It is reasonable to apply the inhibitor in the often used 1.0 M acid solution. Moreover, when working with 1.0 M HCl concentration, it is generally attributed to the fully acidic electrolyte (pH approaches to 0). Since mild steel is highly reactive to HCl, increased concentration might be disastrous and complete dissolution of metal may occur. Therefore, in general practice, the concentration of HCl is 1.0 M. This is a similar test procedure to that of 3.5% NaCl electrolyte aqueous solution. Since the sea

environment contains 3.5% NaCl, it is a standard electrolyte. Similarly, 1.0 M HCl could be used (ASTM International, 2011).

3.2. Weight-Loss studies

Fig. 2 shows the change in corrosion rate and protective efficacy for various immersion times in 1.0 M HCl solution at 303 K at different concentrations of ETO. The rate of corrosion was at its maximum in the uninhibited acid solution, but it was significantly reduced with the addition of the inhibitor. The mild steel sample corroded at a 5.91 mm/yr rate in the absence of an inhibitor, but this decreased to 2.71 mm/yr when 100 ppm ETO was added. The use of 100 ppm ETO resulted in an inhibitory efficiency of 52.6 percent. This finding may be due to the adsorbed inhibitor molecules blocking the corrosion reaction sites, causing the corrosion delay. ETO is a concentration-dependent metabolite. As the concentration of ETO increases, the wear rate decreases while the inhibition efficiency increases. This may be because there is more surface cover (Al-Azawi et al., 2016; Al-Sabagh et al., 2018).

When the ETO concentration is increased from 500 to 600 ppm, the corrosion rate and inhibition efficiency remain almost constant, as shown in Fig. 2. For instance, the inhibition efficiencies of 100, 200, 300, 400, 500 and 600 ppm ETO are 52.6, 77.3, 81.5, 95.2, 98.4 and 98.6% and the corrosion rates are 2.71, 1.74, 1.28, 1.17, 0.97, and 0.94 mm/year, respectively. It appears that when 500 ppm of ETO was applied, the ETO layer covered almost all the active centers and that this addition had only a negligible effect on the efficiency and wear rate.

The inhibitory efficiencies of all tested inhibitor concentrations rise with increasing time from 1 h to 24 h, as shown in Fig. 3. This is because increasing the inhibitor immersion time increases the surface coverage. As the layer covering the surface of mild steel degrades after 24 h, the inhibitory efficiency decreases. Weight loss assessments are particularly valuable in long-term trials because they do not disturb the system significantly, and allow researchers to track the progress of the inhibiting mild steel system over time. The maximum inhibition efficiency of ETO was obtained during the fifth hour, and the inhibitor particles remained on the surface of the mild steel even after the fifth hour, which increased the inhibition efficiency. The ETO molecules lower the corrosive media's dissolving response on the mild steel surface.

Further increase in concentration does not improve inhibition beyond the optimum value. The maximum inhibition efficiency was shown at 500 ppm concentration of ETO and further increasing inhibitor concentration (600 ppm) does not change IE% markedly. Indeed, the corrosion rate values of mild steel decrease on the addition of 100 ppm to 500 ppm of ETO. The increased inhibition efficiency (IE%) and decreased corrosion rate (C_R) might be due to the result of increased adsorption and increased coverage of ETO on the mild steel surface with increasing concentration of ETO. The variation of inhibition efficiency with various immersion times from 1 h to 24 h at 500 ppm ETO is shown in Fig. 2.

The IE% increased to 98% when immersion time increased from 1 h to 5 h, and thereafter no deep change in IE% was noted when immersion time increased from 5 h to 10 h. This is due to the increased corrosion rate of metal with immersion time. It is clear that ETO showed good inhibition for mild steel corrosion in 1.0 M HCl solution.

3.3. Effect of temperature on corrosion rate and inhibition efficiency

The corrosion rate of ETO also was investigated at temperatures of 303, 313, 323, and 333 K. ETO exhibits an increase in corrosion rate with increasing temperature: 1.15 mm/y at 303 K, 1.62 mm/y at 313 K, 1.15 mm/y at 323 K, and 1.82 mm/y at 333 K at an inhibitor concentration of 500 ppm and immersion time of 5 h (Fig. 3). On the other hand, the inhibition efficiency at the same concentrations and immersion time was 98.4% at 303 K, 92.0% at 313 K, 84.8% at 323 K, and 80.1% at 333 K at the inhibitor (Fig. 3). ETO protection efficiency at the inhibition concentration of 600 ppm and exposure time of 5 h are only

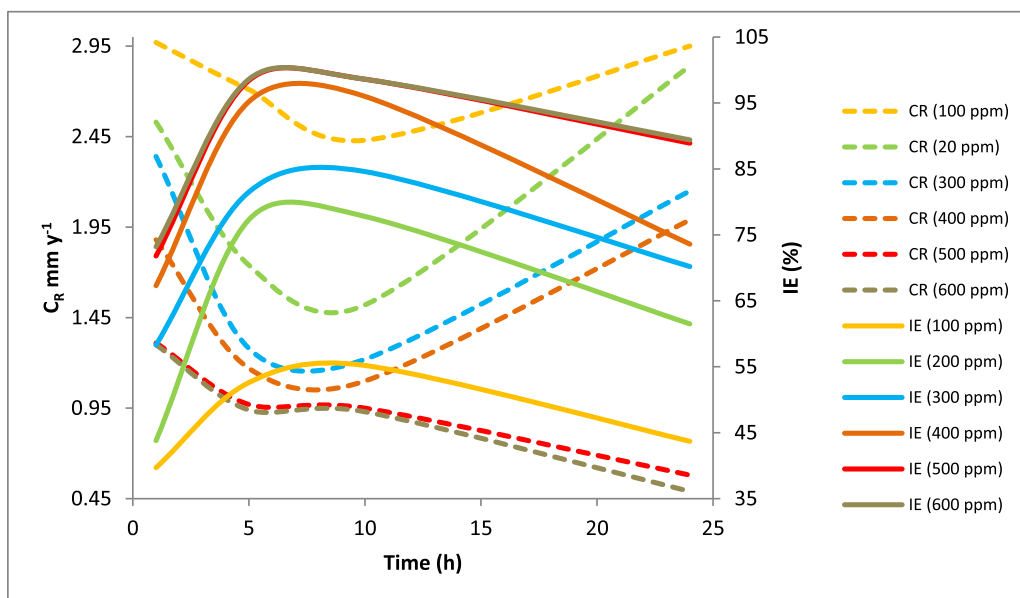


Fig. 2. Corrosion rate and inhibition efficiency for various concentration of ETO of mild steel in 1 M HCl and different immersion times.

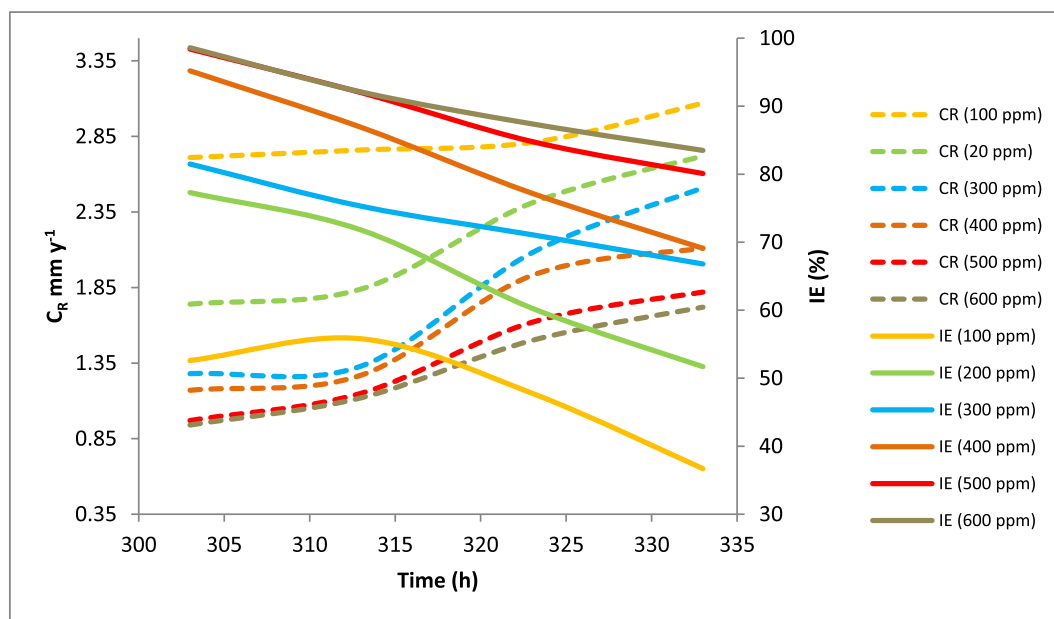


Fig. 3. Corrosion rate and inhibition efficiency for various concentration of ETO of mild steel in 1 M HCl and various Temperatures (303, 313, 323, and 333) for 5 h as immersion time.

83.5% at 333 K, 87.4% at 323 K, 92.1% at 313 K, and 98.6% at 303 K.

Fig. 3 shows that when the temperature rises, the rate of corrosion of ETO rises, while the protection efficacy reduces. This is owing to the belief that ETO chemisorbs on the surface of mild steel (Guo et al., 2017). In this case, the interaction between the inhibitor molecules and the mild steel surface generates coordinating bonds through transfer pairs of electron from the sulfur, oxygen and nitrogen atoms to the unoccupied d-orbitals of iron atoms.

The inhibitory efficiency in hydrochloric acid solution of the current inhibitor “1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO)” can also be evaluated with that of other published inhibitors that are using corrosion inhibitors generated from organic compounds to protect mild steel from corrosion.

Table 2 shows that the examined inhibitor ETO has a significant inhibitory performance when compared to many other manufactured

inhibitors, as well as being a one-shot synthesis. In comparison to the examined inhibitor, the corrosion inhibitors shown in Table 2 had proportionally lower or similar protection performance, thus being significantly costly to synthesize, with processes involving many steps and costly solvents.

3.4. Activation energy (E_a)

Plots of $\log C_R$ against $1/T$ for 100–500 ppm concentration of ETO at 5 h immersion time are shown in Fig. 4. The excellent fit obtained for the points indicates that the experimental findings are consistent. As a result, the slopes derived from the plots are suitable for determining the activation energy of the process for 5 h exposure period.

Tables 3, show the activation energy, E_a for various ETO concentrations and an immersion time of 5 h. The E_a of the inhibited process is

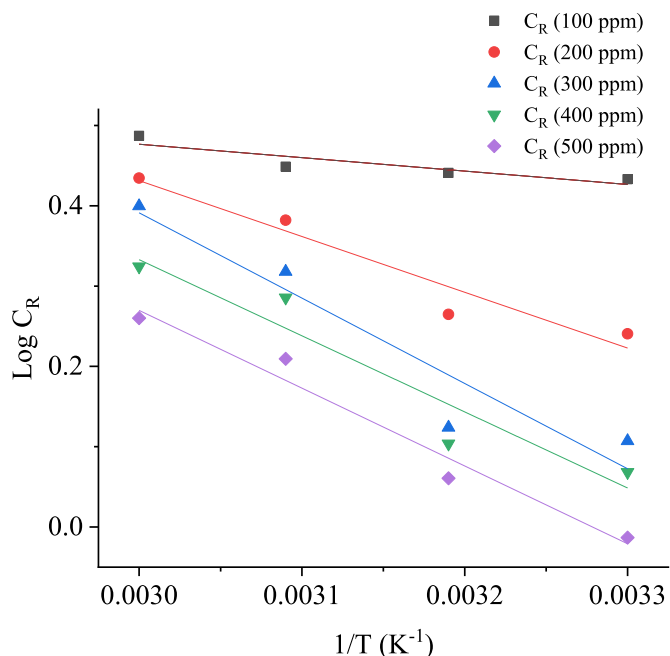
Table 2

Comparison of reported corrosion inhibitors with the one under investigation.

Corrosion inhibitor	Metal	Acid	IE%	Ref.
ETO	MS	HCl	98.4	–
N'-(2-(2-oxomethylpyrrol-1-yl)ethyl)piperidine	MS	HCl	91.9	28
2-Amino-4-phenyl-N-benzylidene-5-(1,2,4-triazol-1-yl)thiazole	MS	HCl	98.1	29
2-amino-4-phenylthiazole	MS	HCl	94.7	29
1-Amino-2-mercapto-5-(4-(pyrrol-1-yl)phenyl)-1,3,4-triazole	MS	HCl	96.3	30
N'-(2-hydroxybenzylidene)-2-(quinolin-8-yl)oxyacetohydrazide	MS	HCl	93.4	31
3-(4-ethyl-5-mercapto-1, 2, 4-triazol-3-yl)-1-phenylpropanone	MS	HCl	97	32
4-ethyl-1-(4-oxo-4-phenylbutanoyl)thiosemicarbazide	MS	H ₂ SO ₄	88.7	33
4-benzyl-1-(4-oxo-4-phenylbutanoyl)thiosemicarbazide	MS	HCl	92.5	34
4-chloro-2-((pyridin-2-ylimino)methyl)phenol	LCS	HCl	92.8	35
2-N-phenylamino-5-(3-phenyl-3-oxo-1-propyl)-1,3,4-oxadiazole	MS	HCl	95.1	36
4-ethyl-1-(4-oxo-4-phenylbutanoyl)thiosemicarbazide	MS	H ₂ SO ₄	88.7	37
4-ethyl-1-(4-oxo-4-phenylbutanoyl)thiosemicarbazide	MS	HCl	96.1	37
4-pyrrol-1-yl-n-(2,5-dimethyl-pyrrol-1-yl)benzoylamine	MS	HCl	95.8	38
N-(1-phenylethylidene)-4-(1H-pyrrol-1-yl)benzohydrazide	MS	HCl	94.5	39
5-((4-fluorobenzylidene)amino)-1,3,4-thiadiazole-2-thiol	MS	HCl	91	40
2-(5-amino-1,3,4-thiadiazol-2-yl)-5-nitrofuranterephthalohydrazide	MS	HCl	83.2	41
isophthalohydrazide	MS	HCl	96.4	42
isophthalohydrazide	MS	HCl	97.2	42
N-(Naphthalen-1-yl)-1-(4-pyridinyl)methanimine	MS	HCl	91.5	43
2-acetylthiophene thiosemicarbazone	MS	HCl	96	44
2-isonicotinoyl-N-phenylhydrazinecarbothioamide	MS	HCl	96.3	45
2-amino-5-(naphthalen-2-ylmethyl)-1,3,4-thiadiazole	LCS	HCl	95.1	46
5-(4-(1H-pyrrol-1-yl)phenyl)-2-mercapto-1,3,4-oxadiazole	MS	HCl	95	47
N-(2,4-dihydroxytoluene)ylidene)-4-methylpyridin-2-amine	LCS	HCl	93.7	48
N-methyl-2-(1-(5-methyl thiophene-2-yl)ethylidene) hydrazine carbothioamide	MS	HCl	95.3	49
1-phenyl-2-(1-phenylethylidene) hydrazine	MS	HCl	83.8	50
1-(1-(4-methoxyphenyl)ethylidene)-2-phenylhydrazine	MS	HCl	95.1	50
2-(2,4-dimethoxybenzylidene)-Nphenylhydrazinecarbothioamide	MS	HCl	94.8	51
2-(5-amino-1,3,4-oxadiazol-2-yl)-5-nitrofuranterephthalohydrazide	MS	HCl	79.4	52
8-piperazine-1-ylmethylumbelliferone	MS	HCl	93.4	53
2-((6-methyl-2-ketoquinoline-3-yl)methylene)hydrazinecarbothioamide	MS	HCl	95.8	54
4-(6-methylcoumarin)acetohydrazide	MS	HCl	94.5	55
4-(Benzoimidazole-2-yl)pyridine	MS	HCl	93.8	56
5,5'-(1,4-phenylene)bis(N-phenyl-1,3,4-thiadiazol-2-amine)	MS	HCl	94	57

*MS= Mild steel; LCS= Low carbon steel.

greater than that of the uninhibited one, as indicated in Table 3. This observation demonstrates that physisorption is used in the adsorption process. The greater energy barrier was responsible for the higher activation energy in the presence of ETO. This occurrence also supports the existence of a complex molecule between the ETO molecules and steel surface (Erna et al., 2019). Because of the increased energy barrier for metal breakdown, these data shows that corrosion inhibition by ETO is possible. The creation of a thin coating on the mild steel surface acts as an energy and mass transfer barrier, raising E_a . As a result, the finding indicates that ETO adsorption on a metal surface is due to physical adsorption. The E_a dropped as the ETO concentration increased, as seen in the Table 3. E_a was, however, still larger than in the uncontrolled solution. E_a was a maximum at 5 h immersion time for ETO

**Fig. 4.** Log C_R versus $1/T$ for various concentrations of ETO at different immersion times.**Table 3**

Thermodynamically parameters at 5 h exposure period of various concentrations of ETO and different solution Temperature.

Concentration of ETO (ppm)	Thermodynamically parameters E_a (KJ/mol)	ΔH (KJ/mol)	ΔS (J/mol/K)
0.00	38.27	–	–
100	77.8	74.9	103
200	81.3	79.2	117.3
300	88.5	86.6	99.1
400	90.3	87.1	71.5
500	89.1	85.9	62.9

concentrations of 100–400 ppm, after which it began to decrease. This result shows that the activation energy grew with the duration of immersion up to 5 h, after which the ability of ETO to create an energy barrier might decrease. Organic inhibitors can lose their inhibiting ability over time, as they are biologically destroyed and interact with the environment, causing them to dissolve in solution. After 10 h, the activation energy of 500 ppm ETO decreases slightly, indicating that the concentration is high enough to counteract the effect of time. It is noticeable that the effect of the activation power at a concentration of 500 ppm is small, with an increase in temperatures from 303 to 333 K.

3.5. Adsorption isotherms

Adsorption isotherms can provide crucial information on the interaction between the inhibitor and the mild steel surface. Various models of the adsorption isotherm are advocated for this method. In the current investigation, we suggested analysis of the adsorption mechanism of ETO molecules using Tempkin, Langmuir, and Freundlich adsorption isotherms, as in Eqs. (5)–(7). The coefficient of correlation (R^2) is employed to evaluate whether the adsorption model is compatible with the experimental findings (Erna et al., 2019).

Based on Fig. 5, the R^2 value of the plot between the natural logarithm of the concentration ($\ln C$ against θ) significantly varies from the unit slope. This indicates that the adsorption of ETO on the mild steel does not obey the Temkin isotherm. Besides, the Langmuir adsorption isotherm model is employed for assessment and is presented in Fig. 6.

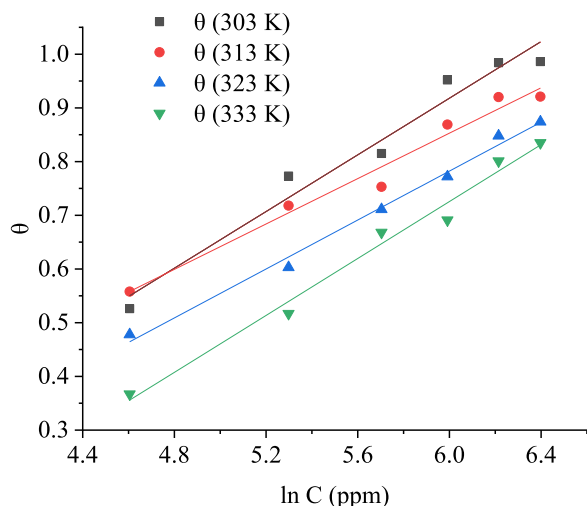


Fig. 5. Temkin isotherm model of various concentrations ETO on the surface of mild steel in 1.0 M HCl at different temperatures.

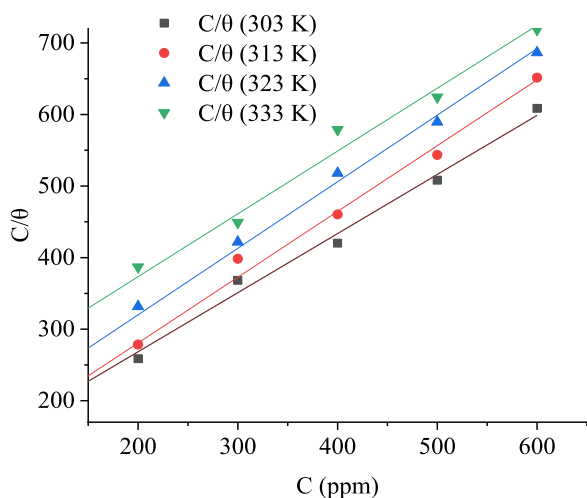


Fig. 6. Langmuir isotherm models of various concentrations of ETO in 1.0 M HCl at different temperatures.

The straight line of C versus C/θ is found with the R^2 value close to 1, and the value of the slope in the Langmuir relation is approximately equal to 1 (Table 3). The above findings confirm that adsorption of ETO on the mild steel surface follows the Langmuir isotherm. The K value is calculated and presented in Table 3. The relation between $\log c$ and $\log \theta$ is related to Freundlich adsorption isotherm model. The correlation coefficient proves that the adsorption of ETO on the mild steel does not follow the Freundlich adsorption isotherm model, as shown in Fig. 7.

3.6. Gibb's free energy

Additionally, the value of ΔG_{ads}° exhibits important data. ΔG_{ads}° is more than -20 kJ mol^{-1} , which suggests that there is a physical interaction between the charged inhibitor and the mild steel surface (physisorption). Whereas if is more negative than -40 kJ mol^{-1} , the adsorption suggests the organic inhibitor molecules transfer onto the metal surface, following to form coordination bonds (chemisorption). Actually, there is no barrier between physisorption and chemisorption in which the physisorption is recognized as the first stage of chemisorption (Abdulsahib et al., 2021; Khudhair et al., 2021; Erna et al., 2019). Consequently, the adsorption of ETO molecules on a mild steel surface obeys physical and chemical adsorption when viewing the ΔG_{ads}° value.

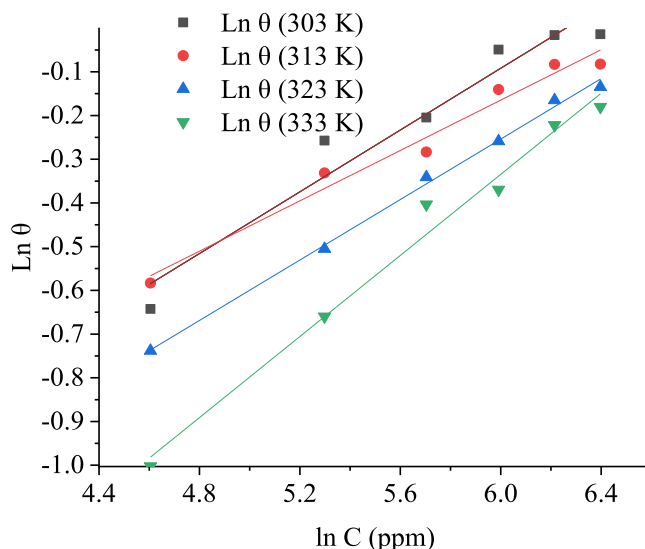


Fig. 7. Freundlich isotherm models of various concentrations of ETO in 1.0 M HCl at different temperatures.

3.7. Density functional theory (DFT)

The adsorption sites of ETO molecules are evaluated through atomic charges (Mulliken charges and Lowdin charges) (Al-Azawi et al., 2016). The atoms with the highest negative charge are the best in coordination with the metal surface and easier to donate pairs of electrons. The data in Tables 4 and 5 indicate that oxygen and nitrogen atoms are positions donating electrons for ETO.

Optimized chemical structure, highest occupied molecular orbital (HOMO), and lowest unoccupied molecular orbital (LUMO) ETO at B3LYP/6–311G(d,p) are demonstrated in Fig. 8. The HOMO and LUMO present the positions in the inhibitor molecules which are donating/accepting electrons (Alkadir Aziz et al., 2021; Alamiery et al., 2021b). According to the highest occupied molecular orbital in Fig. 8, ETO molecules have an ability to donate electrons to the d-orbital of Fe atoms of a mild steel surface at oxygen and nitrogen atoms. On the other hand, the lowest unoccupied molecular orbital suggests the positions in the ETO molecules which receive electrons from Fe atoms.

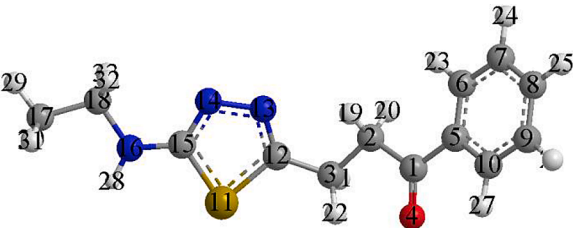
The energies of HOMO (E_{HOMO}) and LUMO (E_{LUMO}), as quantum chemical parameters, in addition to energy gap (ΔE_{L-H}), electronegativity (χ), hardness (η), and softness (S) are determined for ETO (Table 6).

Table 6 shows the quantum chemical properties of the inhibitor. ETO's E_{LUMO} (-5.133 eV) is low, whereas E_{HOMO} (-9.040 eV) is high. As a result, ETO has a great ability to receive electrons. The energy gap (ΔE_{L-H}) is a useful descriptor that reveals the inhibitor's reactivity (adsorption) to the metal surface (Alamiery et al., 2021b). Inhibition efficiency is generally good for organic compounds with a low energy gap value. ETO has a small ΔE_{L-H} value. Furthermore, the values of η , S , and ΔN in Table 6 demonstrate that ETO can better protect the steel surface. This corresponds to the inhibition efficiency experimental finding.

Table 4
Mulliken charges of ETO.

Atom	Charges	Atom	Charges	Atom	Charges
C(1)	0.438272	C(7)	-0.178630	N(13)	-0.323818
C(2)	-0.453021	C(8)	-0.175780	N(14)	-0.369202
C(3)	-0.440774	C(9)	-0.186051	C(15)	0.291944
O(4)	-0.463635	C(10)	-0.155296	N(16)	-0.702669
C(5)	-0.075000	S(11)	0.408500	C(17)	-0.589547
C(6)	-0.201878	C(12)	0.012391	C(18)	-0.186827

Table 5
Lowdin charges of ETO.



Atom	Charges	Atom	Charges	Atom	Charges
C(1)	0.182852	C(7)	−0.094876	N(13)	−0.094234
C(2)	−0.211645	C(8)	−0.078853	N(14)	−0.183070
C(3)	−0.177927	C(9)	−0.094188	C(15)	−0.015154
O(4)	−0.202760	C(10)	−0.062525	N(16)	−0.212759
C(5)	−0.093889	S(11)	0.147805	C(17)	−0.273379
C(6)	−0.079582	C(12)	−0.095664	C(18)	−0.040859

The energy gap, or ΔE_{gap} value, governs chemical reactivity. In terms of reactivity, the higher the inhibition performance molecule, the more reactive it is towards the substrate surface, and the lower the ΔE_{gap} , the more stable it is. As a result, ETO and the Fe substrate form a more stable combination. The dipole moment (μ) is caused by the atoms in the molecule not having a uniform charge distribution. As the dipole moment value decreases, inhibitor molecules accumulate on the Fe surface, increasing ETO's adsorption capacity, which is consistent with the experimental evidence. In the hard-soft acid-base notion, the soft molecule has lower ΔE_{gap} values and higher basicity, and vice versa, when compared to the hard ones. As a result, the soft molecule has adsorption ability due to its easier electron transfer and better corrosion inhibitor than the hard one. According to Lukovit's research, when the number of electrons transmitted (ΔN) is less than 3.6, the inhibition performance as a function of electron transfer improves (Al-Azawi et al., 2016; Alkadir Aziz et al., 2021; Alamiery et al., 2021b).

3.8. A scanning electron microscope (SEM)

The SEM technique was applied to investigate the of mild steel surface characteristics without and with the addition of examined inhibitor. Fig. 9 (a, and b) depicts the modifications that observed in the tested coupons. The surface of mild steel in 1 M hydrochloric acid solution is demonstrated in Fig. 9(a). The coupon surface in corrosive environment having tested inhibitor is demonstrated in Fig. b. The smooth surface of

the Fig. 9(b) was attributable to the inhibitor molecules adsorbing on the coupon surface.

3.9. Mechanism of corrosion inhibition

Because the compound's HOMO orbitals are mostly found in the thiadiazole ring, this ring contains ETO adsorption sites, which can donate electrons to iron's unoccupied d-orbitals. At this point, bonding theories, such as valence bond theory, molecular orbital theory, and crystal field theory, are used to generate iron-inhibitor complexes. The creation of a protective layer that is absorbed on the iron surface is responsible for the inhibitory inertia of organic molecules. In the presence of inhibitors, mild steel corrosion was significantly decreased, according to weight-loss approaches (ETO).

Furthermore, the adsorption isotherm analyses show that the inhibitor's adsorption on the iron surface is well-fitting to the Langmuir adsorption model. Furthermore, the generated protective film's adsorption behavior is principally determined by (1) electrostatic interactions via protonated heteroatoms, and (2) different linkages between inhibitor molecules and the mild steel surface.

The interaction between the inhibitors (ETO) and the mild steel surface is depicted in greater detail in Fig. 10. The interaction between the unpaired electrons and the vacant d-orbital of the Fe atoms is the first route of adsorption between the inhibitor molecule and the mild surface. The second route is the donor-acceptor interactions between the lone electron pairs in the heteroatoms and the unoccupied d-orbital of the iron (mild steel) surface atoms (oxygen, and nitrogen). These active electrons are destined to be shared with the Fe atom's d-orbitals (Lebrini et al., 2007; Redlich and Peterson, 1959).

4. Conclusions

- Synthesized 1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO) has significant corrosion inhibition efficiency for mild steel in 1 M hydrochloric acid solution. This is due to the presence of highly efficient adsorption sites, such as nitrogen and

Table 6
ETO quantum chemical parameters at B3LYP/6–311G(d,p).

E_{HOMO} (eV)	E_{LUMO} (eV)	$\Delta E_{\text{L-H}}$ (eV)	χ (eV)	η (eV)	S (eV^{-1})	μ (D)	ΔN
−9.040	−5.133	3.907	7.0865	1.9535	0.5119	1.9	−0.022

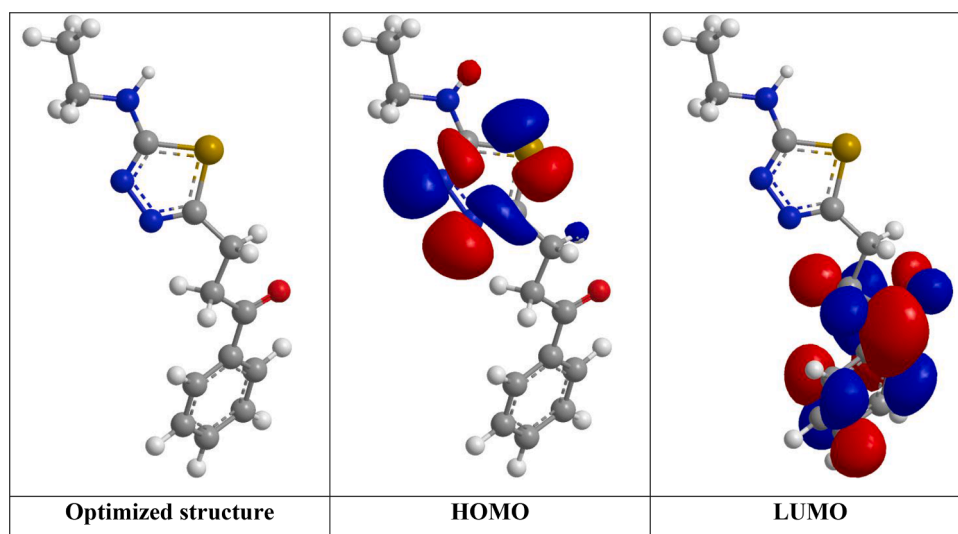


Fig. 8. Optimized structure, HOMO, and LUMO ETO.

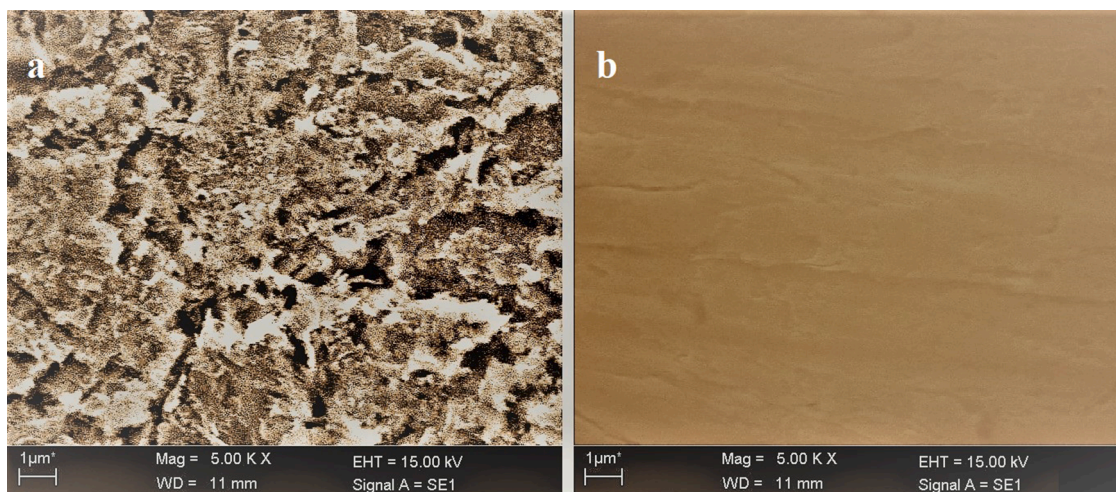


Fig. 9. SEM pictures of mild steel (a) without tested inhibitor, and (b) with tested inhibitor in 1 M HCl solution.

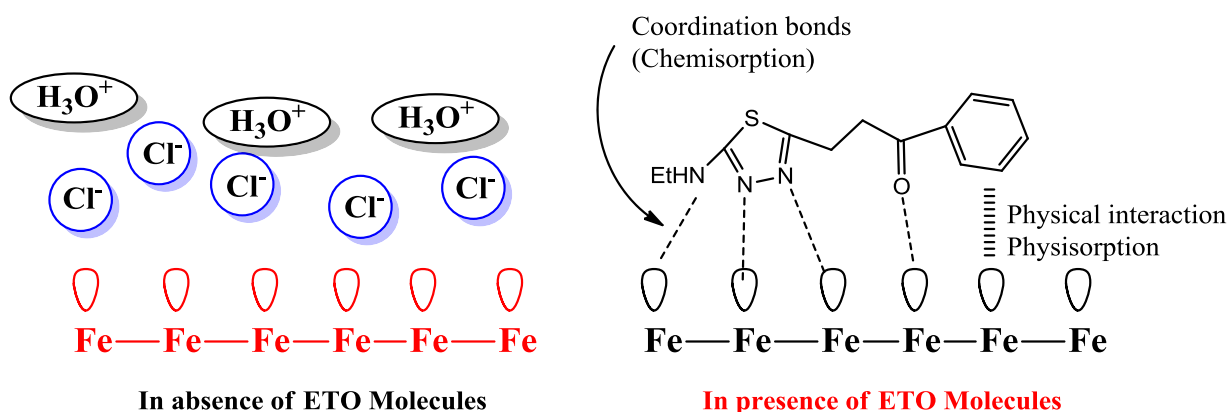


Fig. 10. Simulated corrosion and corrosion inhibition reaction mechanisms of mild steel in 1.0 M hydrochloric acid solution, without, and with, the addition of tested inhibitor.

oxygen atoms, in addition to benzene rings and pi-bonds, which can block the active sites of Fe atoms on the surface of mild steel.

- The weight loss measurements show that the inhibition efficiency values increased with increase in inhibitor concentration but decreased with increase in temperature and the highest inhibition efficiency at the optimum inhibitor concentration was above 98%
- DFT studies demonstrate that the energy of adsorption of the ETO molecules over the mild steel was more than that of the chloride and hydronium ions, and agrees with the decrease in corrosion rate due to the adsorption of inhibitor over the mild steel surface in place of chloride and hydronium ions.
- The adsorbed layer formed on the mild steel surface was characterized by SEM technique.

Declaration of competing interest

The corrosion inhibition potency of 1-(2-ethylamino-1,3,4-thiadiazol-5-yl)-3-phenyl-3-oxopropan (ETO) for mild steel in 1.0 M hydrochloric was investigated by applying the weight loss techniques, and quantum chemical calculations. Conditions that determine the potential of corrosion inhibition including concentration, the structure of ETO, and solution temperature were chosen for the study. The highest inhibitive efficacy of ETO is 98.4% at the optimum concentration (500 ppm) and room Temperature. In fact, ETO has superior corrosion inhibitive potency in the HCl environment owing to the presence of the thiaziazole and benzene rings in addition to carbonyl and amino groups.

Applicable thermodynamical equations were used to calculate the activation energy, enthalpy, and entropy. The adsorption isotherms were applied to determine the Gibbs free energy difference. View from the experimental findings of the investigation revealed that the activation energy of an inhibited process was higher when compared with the uninhibited one. Furthermore, the greater inhibition efficiency with improving temperature and the values of ΔG^0 indicated that ETO molecules filmed the mild steel surface in both chemical adsorption and physical interactions. The adsorption process on the mild steel surface obeys Langmuir adsorption isotherm. Both experimental and DFT findings in the current investigation are in excellent agreement.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.sajce.2022.06.011>.

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